

Design and Development of an Intelligent Data Processing Pipeline for High-Energy Astrophysical Signals from Satellite-Based Observations

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In [ ]: # pip install numpy pandas matplotlib astropy scikit-learn
#
# conda create -n satdata python=3.11 scikit-learn pandas matplotlib ipyk
# conda activate satdata
# conda deactivate
```

```
In [1]: # Satellite Data / Dataset

import numpy as np

# segnale con rumore
time = np.linspace(0, 10, 1000)
signal = np.sin(time)

# aggiungo rumore
noise = np.random.normal(0, 0.5, 1000)
data = signal + noise

# aggiungo un "evento"
data[400:410] += 5

print(data)
```

[-3.83003718e-01	4.83092542e-01	-1.17938262e-01	2.44095354e-01
4.18333774e-01	-1.48630314e-01	-5.35398005e-01	5.28307550e-01
-1.10070906e-01	2.99909091e-01	-9.76971184e-02	4.23781272e-01
-3.48990083e-01	-6.61336757e-01	4.65868527e-01	2.45647423e-01
5.56981792e-01	-3.02769124e-01	3.09810067e-01	5.95748474e-01
2.23331111e-02	1.32470160e+00	5.58574101e-01	8.64066207e-01
4.46215315e-01	2.13074001e-01	2.38957788e-01	-2.07011178e-01
3.34819134e-01	-1.03762983e+00	5.08220632e-01	-6.49387365e-01
6.16931069e-01	5.08625507e-01	-1.18413272e-01	5.21390146e-01
-6.14653739e-01	1.15248086e+00	1.41315011e-01	1.06790944e+00
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1.25873292e+00	1.70242280e+00	5.11864542e-02	4.74956331e-01
1.04929179e+00	1.14587339e-01	1.43541806e+00	1.77759474e+00
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9.97935542e-01	7.76344187e-01	1.05018647e+00	6.73470427e-01
6.07927066e-01	3.05189607e-01	2.99402945e-01	7.87723094e-01
5.60385142e-01	8.85036539e-01	-1.39752426e-01	-1.00894338e-02
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9.17537273e-01	1.10726320e+00	1.29195083e+00	1.63613640e+00
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8.41179519e-01	1.47641973e+00	8.94653059e-01	1.64139153e+00
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1.22887791e+00	7.56623700e-01	3.67606828e-01	1.08486366e+00
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1.07751525e+00	9.68311525e-01	2.04533658e+00	1.17778926e+00
1.69754400e+00	9.56383164e-01	8.67488552e-01	1.16296630e+00
1.56601779e+00	7.17612051e-01	-3.19120831e-03	1.35030063e+00
1.18711347e+00	7.12749069e-01	2.23839918e-01	4.63853163e-02
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1.54335461e+00	1.62593521e+00	5.98381748e-01	2.31820342e+00
1.86310576e+00	1.40933927e+00	6.54920365e-01	1.88775442e+00
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1.02907838e+00	7.71093612e-01	8.34655858e-01	9.79070416e-01
4.09307302e-01	5.86927623e-02	5.71819433e-01	5.05293839e-01
4.89997543e-01	1.96243536e-01	8.60866814e-01	1.14267952e+00
1.41335500e+00	1.13776538e+00	3.52753921e-01	1.17598566e+00
1.51249404e+00	1.36904843e+00	6.68828851e-01	1.37082535e+00
5.63123960e-01	5.45168215e-01	9.50729813e-01	5.32667745e-01
1.06056647e+00	2.05708148e-01	1.10688755e+00	1.63889203e+00
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4.55368294e-01	3.52950615e-01	5.46984473e-01	9.86869545e-01
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2.05069555e+00	1.42835139e+00	-1.04447181e-01	1.37653954e+00
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1.15844001e+00	8.41587637e-01	1.72228358e+00	7.99372702e-01
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3.11532386e-01	1.64947824e+00	9.72565634e-01	2.57272325e-02
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1.50357169e+00	7.92410718e-01	6.54737706e-01	2.78579673e-01
2.56578002e-01	1.43551627e+00	1.11442615e+00	6.33326195e-01
5.58546123e-01	1.54040830e+00	9.79033732e-01	7.83015543e-01
1.35551700e+00	3.60174563e-01	6.97959878e-01	8.08375866e-01
1.10716733e+00	3.74436936e-01	1.13539125e+00	2.42724167e-01
4.62098007e-01	7.10045119e-01	1.40323643e+00	2.86746255e-01
7.76383860e-01	1.21993804e+00	3.32043438e-01	6.77877541e-01
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2.97539807e-01	-4.51701385e-01	7.80462903e-01	4.76677912e-01
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1.07208985e+00	3.66165252e-01	3.95535716e-01	-6.17069406e-01
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-3.78075749e-02	1.52981436e-01	1.30696774e+00	5.65879658e-01
1.08767920e+00	-4.96327063e-01	-1.37266759e-01	-3.03236594e-01
-3.81251371e-01	1.28017090e-01	1.06641392e-02	9.31013748e-02
9.61986746e-01	-3.92885664e-01	-1.76225410e-01	-3.86131213e-01
1.69990082e-01	-4.98722480e-01	-1.24334138e+00	-7.96201430e-01
2.34198014e-01	-2.51409153e-01	-5.79902067e-01	3.46778239e-01

```

-2.29388052e-01 -5.04488411e-01 -8.34173192e-02 -1.09706354e+00
-5.66851630e-01 -8.73903002e-01 -1.04754483e+00 -3.05016165e-01
-4.78688249e-01 9.12406173e-02 -2.96105154e-01 -6.84549627e-01
4.42140900e-03 -5.17034647e-01 -1.07649532e+00 7.81471391e-01
1.69438055e-01 4.94083601e-02 -1.07695340e+00 2.06780389e-01
-8.84744778e-01 -9.01785082e-01 -8.80680075e-01 3.27266090e-01
-6.43371852e-01 2.00123565e-01 -7.99952192e-01 -3.02711545e-01
2.44254317e-01 8.88559387e-02 -6.33978018e-01 -3.94778908e-01
-5.75277248e-01 -6.84840773e-01 -3.06881891e-01 1.24855981e-01
-7.18454257e-02 -8.79861066e-01 -6.01912825e-01 2.52531843e-01]

```

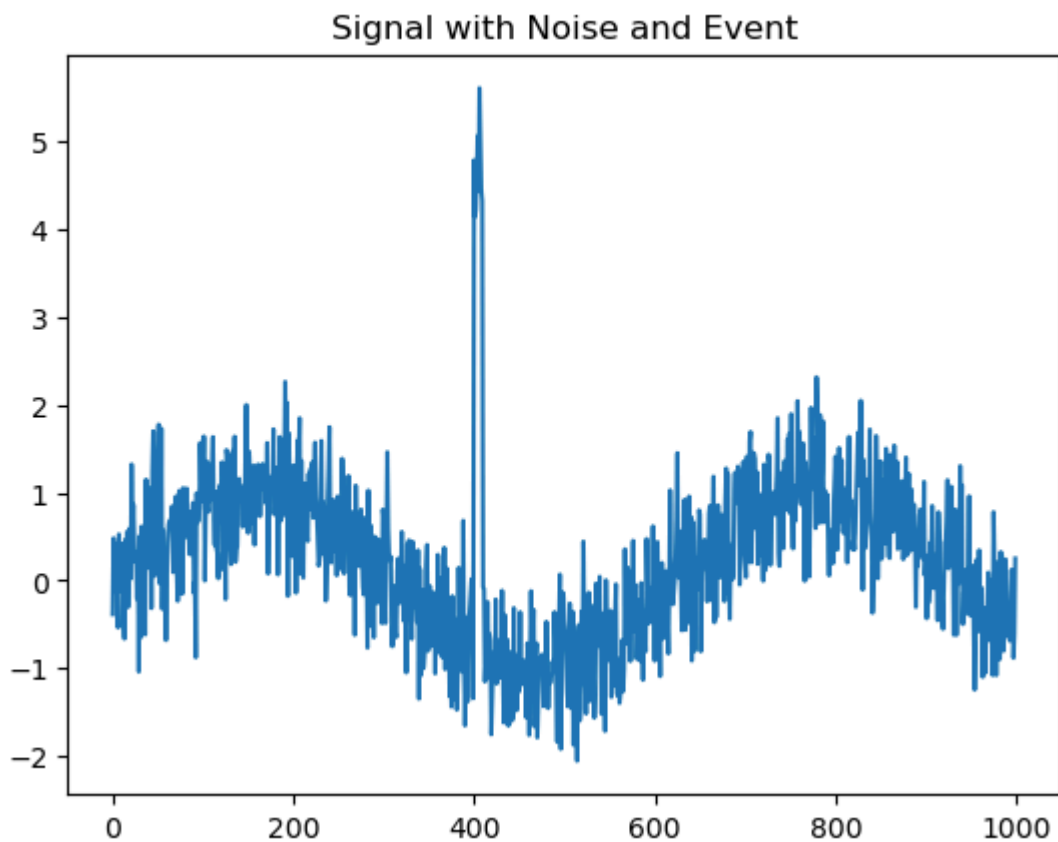
```

In [2]: # Signal Processing

import matplotlib.pyplot as plt

plt.plot(data)
plt.title("Signal with Noise and Event")
plt.show()

```



```

In [3]: # Event Detection (ML)

threshold = 3

events = np.where(data > threshold)[0]

print(events[:10])

[400 401 402 403 404 405 406 407 408 409]

```

```

In [4]: # Anomaly Detection (AI Agent Layer)

from sklearn.ensemble import IsolationForest

model = IsolationForest(contamination=0.01)

```

```
data_resaped = data.reshape(-1, 1)

model.fit(data_resaped)
pred = model.predict(data_resaped)

anomalies = np.where(pred == -1)[0]

print(anomalies[:10])
```

[400 401 402 403 404 405 406 407 408 409]

In [5]: *# Database (MongoDB)*

```
events_list = [{"index": int(i), "value": float(data[i])} for i in events]

print(events_list[:5])
```

[{'index': 400, 'value': 4.789672375970709}, {'index': 401, 'value': 4.147233060893527}, {'index': 402, 'value': 4.21916299006023}, {'index': 403, 'value': 4.590785922715513}, {'index': 404, 'value': 5.065907895826223}]

In [7]: *# Dashboard (Streamlit)*